

Introduction to Probabilistic Machine Learning

Bayesian Classification

Ralf Herbrich

Overview

1. Bayesian Classification Learning
2. Bayesian Classification Learning via Approximate Message Passing

**Introduction to
Probabilistic Machine
Learning**

*Unit 9 – Bayesian
Classification*

Overview

1. **Bayesian Classification Learning**
2. Bayesian Classification Learning via Approximate Message Passing

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Bayesian Classification Learning Setting

■ Given:

1. **Training Data:** $D \in (\mathcal{X} \times \{-1, +1\})^n$ of n (labelled) examples from input space $\mathcal{X}$
2. **Linear Basis Functions:** Basis function mapping $\phi: \mathcal{X} \rightarrow \mathbb{R}^M$ and linear function model

$$f(x; \mathbf{w}) := \mathbf{w}^T \phi(x)$$

3. **Likelihood of functions:** weight vector feature vector

$$p(D|f) = p(D|\mathbf{w}) = \prod_{i=1}^n \int_0^{+\infty} \mathcal{N}(t_i; y_i \mathbf{w}^T \phi(x_i), \beta^2) dt_i \text{Ber}\left(1; g\left(\frac{y_i \mathbf{w}^T \phi(x_i)}{\beta}\right)\right)$$

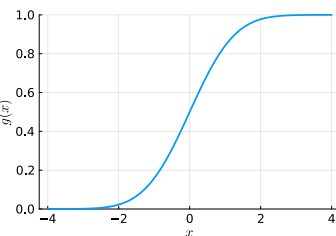
4. **Prior belief over functions:**

$$p(f) = p(\mathbf{w}) = \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

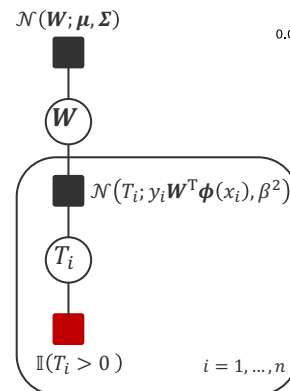
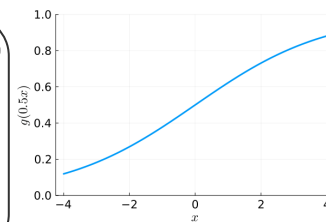
- ## ■ Bayesian Inference: Posterior belief over functions:

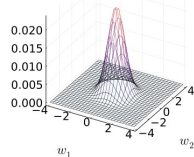
$$p(f|D) = p(\mathbf{w}|D) = \frac{\prod_{i=1}^n \text{Ber}\left(1; g\left(\frac{y_i \mathbf{w}^T \phi(x_i)}{\beta}\right)\right) \cdot \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma})}{\int_{\mathbb{R}^M} \prod_{i=1}^n \text{Ber}\left(1; g\left(\frac{y_i \tilde{\mathbf{w}}^T \phi(x_i)}{\beta}\right)\right) \cdot \mathcal{N}(\tilde{\mathbf{w}}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) d\tilde{\mathbf{w}}}$$

$$g(t) = \int_0^{\infty} \mathcal{N}(z; t, 1) dz$$



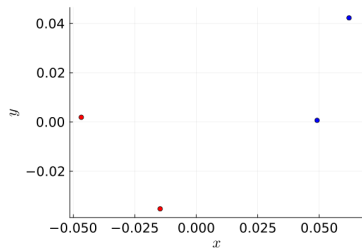
$$\lim_{s \rightarrow \infty} g(s \cdot t) = \mathbb{I}(t > 0)$$





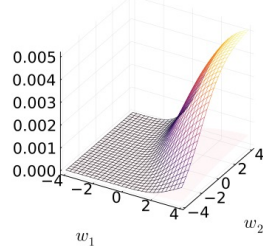
Bayesian Classification Learning in Pictures

Input Data

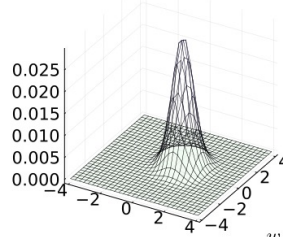


$n = 4$

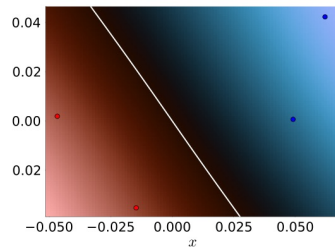
Likelihood



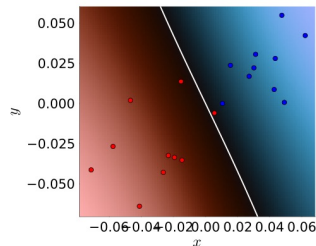
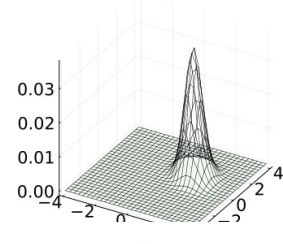
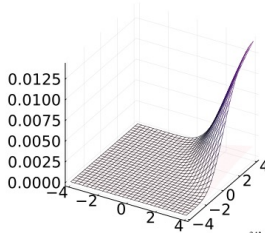
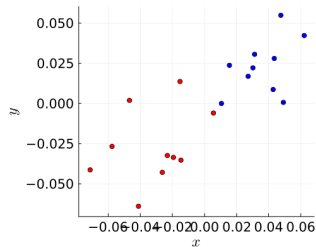
Posterior



Predictive Distribution

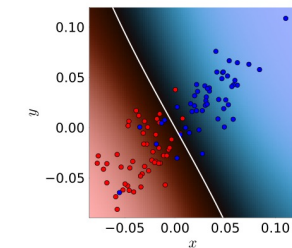
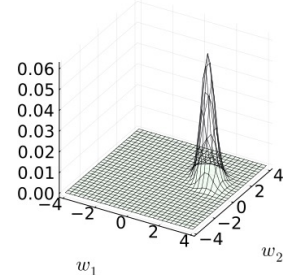
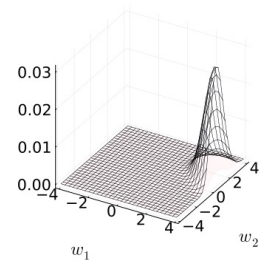
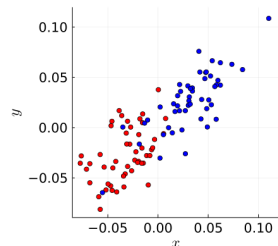


$n = 20$



ine

$n = 100$



Overview

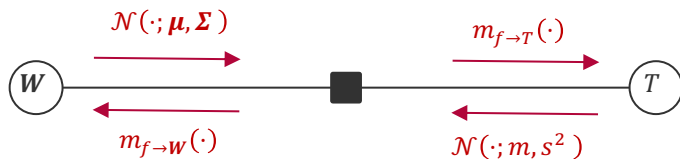
1. Bayesian Classification Learning
- 2. Bayesian Classification Learning via Approximate Message Passing**

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Update Equations for Combined Factors

Gaussian Projection-Noise Factor $\mathcal{N}(T; W^T x, \beta^2)$

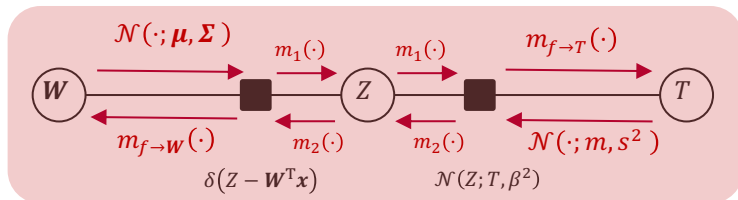


$$m_{f \rightarrow T}(t) = \mathcal{N}(t; \mu^T x, x^T \Sigma x + \beta^2)$$

$$m_{f \rightarrow W}(w) = \mathcal{G}\left(w; \frac{m}{s^2 + \beta^2} x, \frac{1}{s^2 + \beta^2} x x^T\right)$$

$$m_{f \rightarrow T}(t) = \int \mathcal{N}(t; w^T x, \beta^2) \cdot \mathcal{N}(w; \mu, \Sigma) dw = \int \left[\int \delta(z - w^T x) \cdot \mathcal{N}(w; \mu, \Sigma) dw \right] \cdot \mathcal{N}(t; z, \beta^2) dz$$

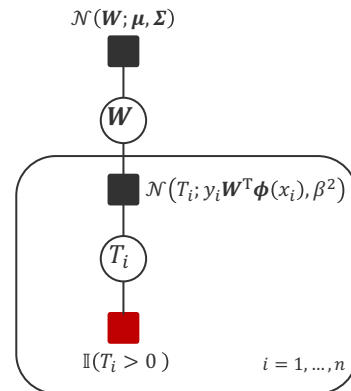
$$m_{f \rightarrow W}(w) = \int \mathcal{N}(t; w^T x, \beta^2) \cdot \mathcal{N}(t; m, s^2) dt = \int \left[\int \mathcal{N}(t; z, \beta^2) \cdot \mathcal{N}(t; m, s^2) dt \right] \cdot \delta(z - w^T x) dz$$



$$m_1(z) = \mathcal{N}(z; \mu^T x, x^T \Sigma x)$$

$$m_2(z) = \mathcal{N}(z; m, s^2 + \beta^2)$$

Helper factor tree to derive closed form message equations



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Unit 9 – Bayesian Classification

Bayesian Linear Classification by Message Passing

- **Factor Tree:** Each training example is summarized in an M -dimensional message

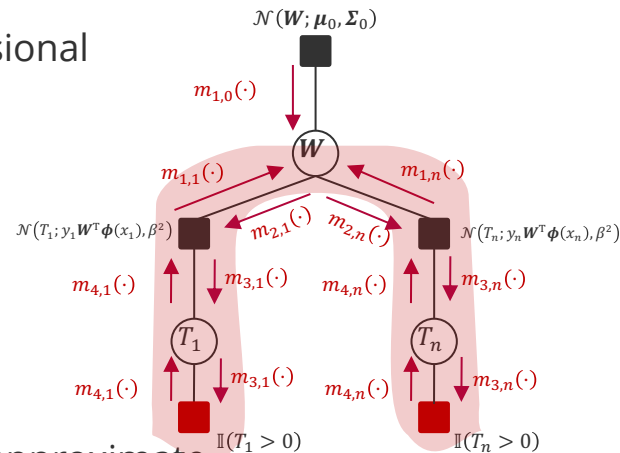
- Prior Message $m_{1,0}(\mathbf{w}) = \mathcal{G}(\mathbf{w}; \boldsymbol{\Sigma}_0^{-1} \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0^{-1}) = p(\mathbf{w})$
- Data Message $m_{1,i}(\mathbf{w}) = \mathcal{G}(\mathbf{w}; a_i y_i \boldsymbol{\phi}(x_i), b_i y_i^2 \boldsymbol{\phi}(x_i) \boldsymbol{\phi}^T(x_i)) = p(y_i | \mathbf{w})$

- **Posterior:** Multiplying prior and data messages we have

$$p(\mathbf{w} | D) = \mathcal{G}\left(\mathbf{w}; \boldsymbol{\Sigma}_0^{-1} \boldsymbol{\mu}_0 + \sum_{i=1}^n a_i y_i \boldsymbol{\phi}(x_i), \boldsymbol{\Sigma}_0^{-1} + \sum_{i=1}^n b_i \boldsymbol{\phi}(x_i) \boldsymbol{\phi}^T(x_i)\right)$$

- **Iterative Approximations:** Since each data message involves an approximate factor, we need to iterate message passing until convergence!

- We maintain the (approximate) posterior $p(\mathbf{w} | D) = \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ and the (a_i, b_i)
- 1. Compute $m_{2,i}(\cdot)$ in scale-location parameters (using Sherman-Morrison formula)
- 2. Compute $m_{3,i}(\cdot)$ in scale-location parameters by Gaussian projection-noise factor
- 3. Compute $m_{4,i}(\cdot)$ in natural parameters by matching moments
- 4. Update $p(\mathbf{w} | D) = \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ in scale-location parameters (using Sherman-Morrison)

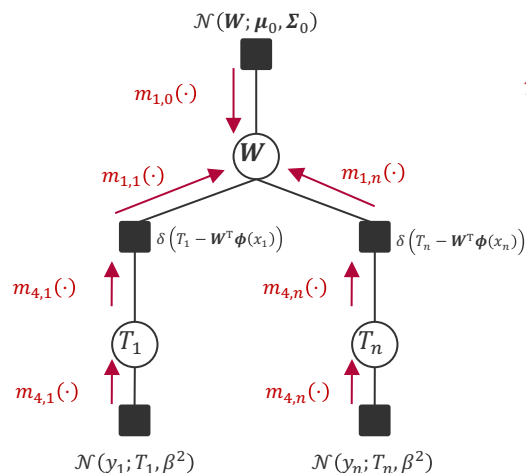


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Relation of Bayesian Linear Classification to Regression

Linear Regression

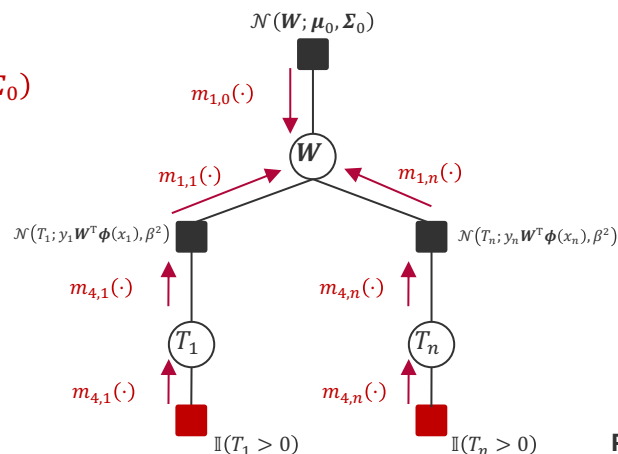


$$m_{1,0}(w) = \mathcal{N}(w; \mu_0, \Sigma_0)$$

$$m_{1,i}(w) = \mathcal{G}(w; \beta^{-2} y_i \phi(x_i), \beta^{-2} \phi(x_i) \phi^T(x_i))$$

The a_i and b_i are effectively
precision-mean and precision of the message $m_{4,i}(\cdot)$!

Linear Classification



$$m_{1,i}(w) = \mathcal{G}(w; a_i y_i \phi(x_i), b_i \phi(x_i) \phi^T(x_i))$$

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Matrix Inverse of a Rank-1 Update

- **Naïve Inverse:** For any invertible matrix $A \in \mathbb{R}^{n \times n}$ we can use the Cholesky decomposition $A = LL^T$ and solve

$$L^T \setminus (L \setminus e_i) = a_i^{-1} \longleftarrow \mathcal{O}(n^3) \text{ over } i = 1, \dots, n$$

to compute the i th column of the inverse A^{-1} .

- **Rank-1 Update:** Given two vectors $u \in \mathbb{R}^n$ and $v \in \mathbb{R}^n$, a rank-1 update is

$$A + uv^T$$

- **Problem:** Is there a computationally efficient way to compute $(A + uv^T)^{-1}$ from A^{-1} ?
- **Why?** Because there are only $2n$ additional parameters rather than $\mathcal{O}(n^2)$!

- **Sherman-Morrison Formula:** For any invertible matrix A it holds that

$$(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u}$$

$\mathcal{O}(n^2)$ (pointing to A^{-1}) $\mathcal{O}(n^2)$ (pointing to $A^{-1}uv^T A^{-1}$) $\mathcal{O}(n^2)$ (pointing to $v^T A^{-1}u$)

- **Proof:**

$$(A + uv^T)^{-1}(A + uv^T) = \left(A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u} \right) (A + uv^T) = I + \frac{A^{-1}uv^T \cdot (1 + v^T A^{-1}u) - A^{-1}uv^T - A^{-1}uv^T A^{-1}uv^T}{1 + v^T A^{-1}u} = I$$

Steps 1 & 4: Update Data Message and Marginal

- Rank-1 Updates of Σ and μ with addition of $\Delta_i^a \cdot y_i \phi_i$ and $\Delta_i^b \cdot \phi_i \phi_i^T$

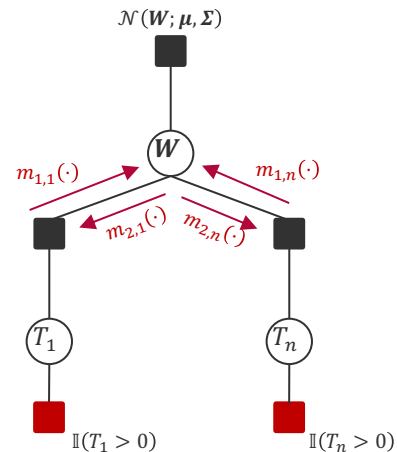
Precision matrix

No inverse needed but just matrix multiplications!

$$\Sigma_{\text{new}} = (\Sigma^{-1} + \Delta_i^b \phi_i \phi_i^T)^{-1} = \Sigma - \frac{\Delta_i^b}{1 + \Delta_i^b \cdot \phi_i^T \Sigma \phi_i} \cdot (\Sigma \phi_i)(\Sigma \phi_i)^T$$

$$\mu_{\text{new}} = (\Sigma^{-1} + \Delta_i^b \phi_i \phi_i^T)^{-1} (\Sigma^{-1} \mu + \Delta_i^a y_i \phi_i) = \mu + \Sigma \phi_i \left(\frac{\Delta_i^a \cdot y_i - \Delta_i^b \cdot \phi_i^T \mu}{1 + \Delta_i^b \cdot \phi_i^T \Sigma \phi_i} \right)$$

$$\mu + \Delta_i^a y_i \Sigma \phi_i - \frac{\Delta_i^b}{1 + \Delta_i^b \cdot \phi_i^T \Sigma \phi_i} \cdot (\Sigma \phi_i) \phi_i^T \mu - \frac{\Delta_i^b \Delta_i^a y_i}{1 + \Delta_i^b \cdot \phi_i^T \Sigma \phi_i} \cdot (\Sigma \phi_i) \phi_i^T \Sigma \phi_i$$



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- Step 1: Compute $m_{2,i}(\mathbf{w})$ by removing i th data factor using $\Delta_i^a = -a_i$ and $\Delta_i^b = -b_i$
- Step 4: Update μ and Σ of $p(\mathbf{w}|D)$ using $\Delta_i^a = a_i^{\text{new}} - a_i$ and $\Delta_i^b = b_i^{\text{new}} - b_i$

Bayesian Linear Classification: Update of i^{th} Data Message

- **Step 2:** $m_{3,i}(t_i) = \mathcal{N}(t_i; m_i, s_i^2)$ follows from the Gaussian projection-noise factor

$$m_i = \frac{y_i \phi_i^T \mu - a_i \phi_i^T \Sigma \phi_i}{1 - b_i \phi_i^T \Sigma \phi_i} \quad s_i^2 = \frac{\phi_i^T \Sigma \phi_i}{1 - b_i \cdot \phi_i^T \Sigma \phi_i} + \beta^2$$

- **Step 3:** $m_{4,i}(t_i) = \mathcal{G}(t_i; \tau_i, \rho_i)$ follows from the greater-than factor

Mean correction
of truncated Gaussian

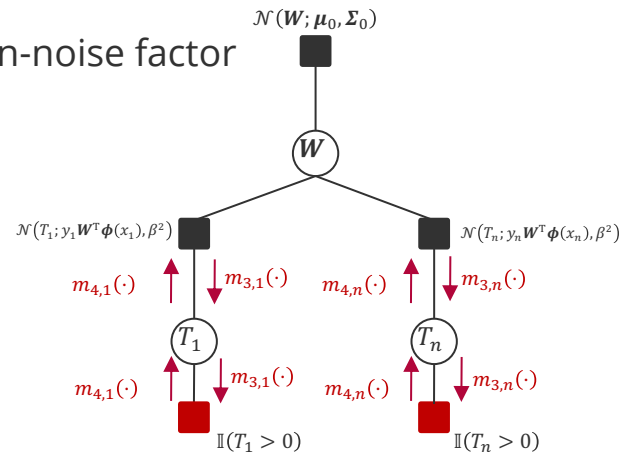
$$\tau_i = \frac{s_i \cdot v\left(\frac{m_i}{s_i}\right) + m_i \cdot w\left(\frac{m_i}{s_i}\right)}{s_i^2 \cdot \left(1 - w\left(\frac{m_i}{s_i}\right)\right)}$$

Variance correction
of truncated Gaussian

$$\rho_i = \frac{w\left(\frac{m_i}{s_i}\right)}{s_i^2 \cdot \left(1 - w\left(\frac{m_i}{s_i}\right)\right)}$$

- **Final Update**

$$a_i^{\text{new}} = \frac{\tau_i}{1 + \rho_i \cdot \beta^2} \quad b_i^{\text{new}} = \frac{\rho_i}{1 + \rho_i \cdot \beta^2}$$



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Bayesian Linear Classification: Putting It all Together

Given: n training examples (x_i, y_i) where $y_i \in \{-1, +1\}$ and feature map $\phi: \mathcal{X} \rightarrow \mathbb{R}^M$

1. **Initialize** $\phi_i := \phi(x_i)$, $\Sigma = \lambda^2 I$, $\mu = \mathbf{0}$ and $a_i = b_i = 0$
2. **Repeat until** maximum change component-wise in $\mathbf{a}$ and $\mathbf{b}$ is small (e.g., $\leq 10^{-6}$)

For all $i = 1, \dots, n$

Step 1: Compute $\mathbf{z} = \Sigma \phi_i$, $e = \phi_i^T \mathbf{z}$, $f = \phi_i^T \mu$, $c = 1 - b_i \cdot e$

Step 2: Compute $m = \frac{y_i \cdot f - a_i \cdot e}{c}$ and $s^2 = \frac{e}{c} + \beta^2$

Step 3: Compute $\tau = \frac{s \cdot V + m \cdot W}{s^2 \cdot (1 - W)}$ and $\rho = \frac{W}{s^2 \cdot (1 - W)}$ for $V = v\left(\frac{m}{s}\right)$ and $W = w\left(\frac{m}{s}\right)$

Compute $a_i^{\text{new}} = \frac{\tau}{1 + \rho \cdot \beta^2}$ and $b_i^{\text{new}} = \frac{\rho}{1 + \rho \cdot \beta^2}$

Step 4: Compute $d = 1 + (b_i^{\text{new}} - b_i) \cdot e$

Update $\Sigma \leftarrow \Sigma - \frac{b_i^{\text{new}} - b_i}{d} \cdot \mathbf{z} \mathbf{z}^T$ and $\mu \leftarrow \mu + \frac{y_i \cdot (a_i^{\text{new}} - a_i) - f \cdot (b_i^{\text{new}} - b_i)}{d} \cdot \mathbf{z}$

Update $a_i \leftarrow a_i^{\text{new}}$ and $b_i \leftarrow b_i^{\text{new}}$

End For

Majority of algorithm is additions, multiplication, and division of scalars!

Algorithm requires **no** matrix inverses – just matrix products!

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Unit 9 – Bayesian Classification

- $$p(\mathbf{w}|D) = \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

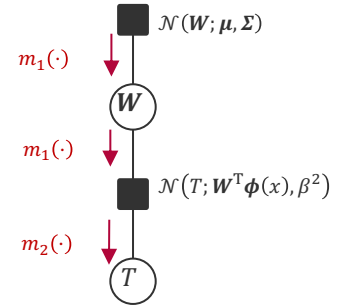
- Prediction at new test point x is proportional to $m_2(\cdot)$
- Posterior $p(\mathbf{w}|D)$ is the message $m_1(\cdot)$ to the Gaussian noise projection factor at the test point x $m_{2,1}$
- To avoid recomputing this message for every test point x we simply store the message $m_1(\mathbf{w}) = p(\mathbf{w}|D)$ as the “model” $m_{2,1}$



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- **Prediction Tree:** Simple factor chain given posterior $p(\mathbf{w}|x, D) = \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$

- Posterior Message $m_1(\mathbf{w}) = \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = p(\mathbf{w}|x, D)$
- Projection-Noise Message $m_2(t) = \mathcal{N}(t; \boldsymbol{\mu}^T \boldsymbol{\phi}(x), \boldsymbol{\phi}^T(x) \boldsymbol{\Sigma} \boldsymbol{\phi}(x) + \beta^2) = p(t|x, D)$
- Outcome Probability $\int_0^\infty m_2(t) dt = p(y = +1|x, D)$



- **Bayesian Linear Probit Regression in Matrix Notation**

$$p(y = +1|x, D) = \int_0^{+\infty} \mathcal{N}(t; \boldsymbol{\mu}^T \boldsymbol{\phi}(x), \beta^2 + \boldsymbol{\phi}^T(x) \boldsymbol{\Sigma} \boldsymbol{\phi}(x)) dt$$

data uncertainty

model uncertainty

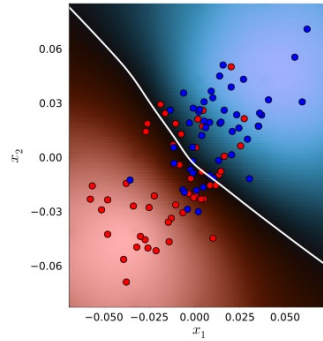
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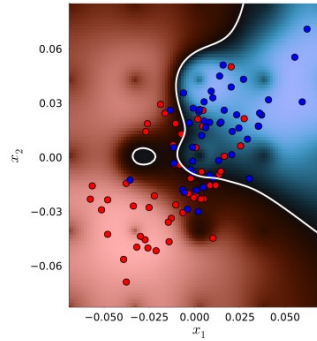
Bayesian Linear Classification in Pictures

$n = 100$

$\lambda = 1$

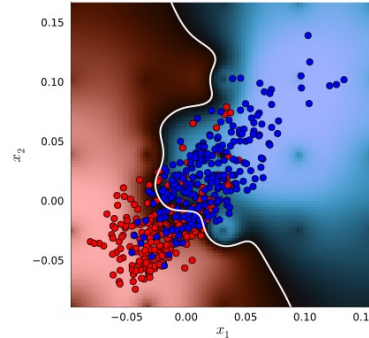
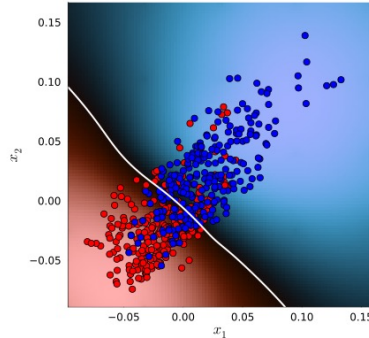


$\lambda = 10$



225 features of
value of 225 Gaussian
density in 2D on a 5x5 grid
at 9 resolutions

$n = 500$



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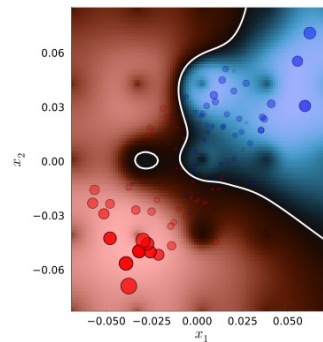
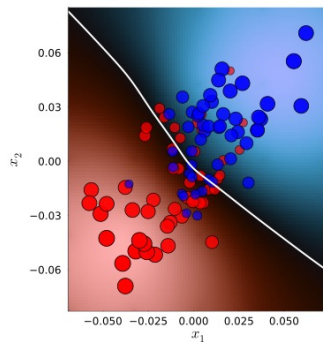
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Bayesian Linear Classification in Pictures (ctd)

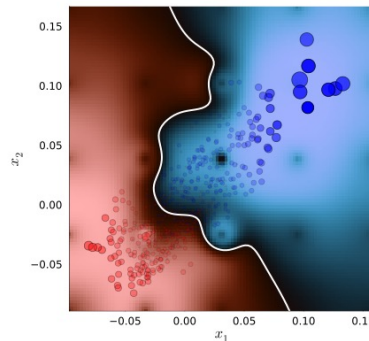
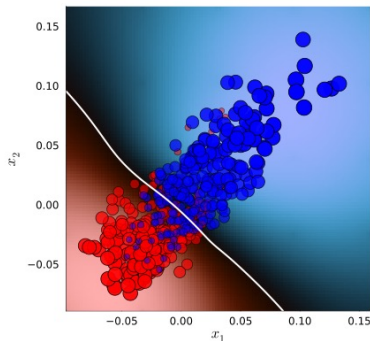
$\lambda = 1$

$\lambda = 10$

$n = 100$



$n = 500$



$$\text{Size} \propto \frac{a_i}{b_i} = \mu_i$$

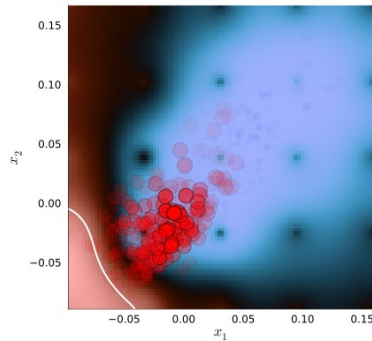
$$\text{Transparency} \propto \sqrt{\frac{1}{b_i}} = \sigma_i$$

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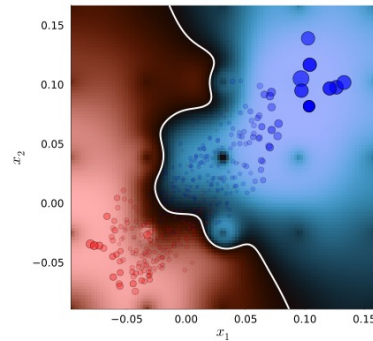
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Bayesian Linear Classification Convergence Pictures

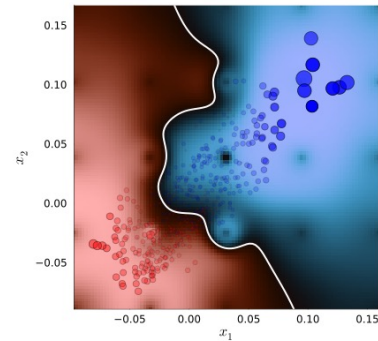
Iteration 1



Iteration 2



Iteration 3



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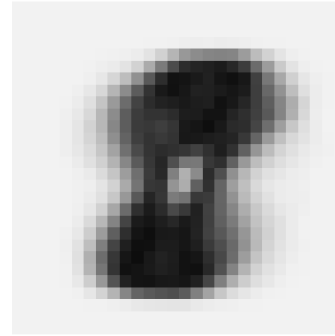
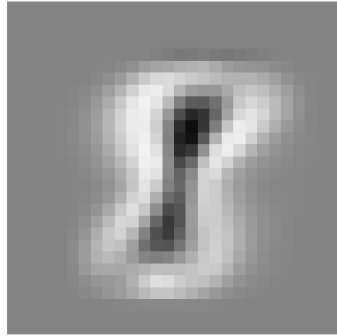
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Bayesian Linear Classification for MNIST

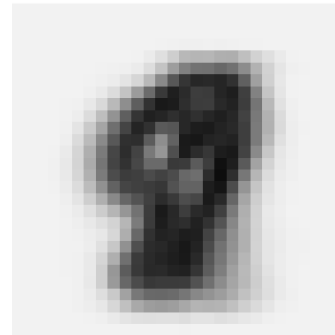
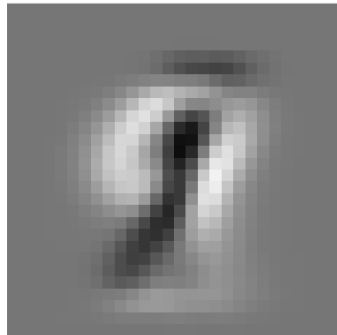
μ

$\text{diag}(\Sigma)$

“1” vs. “8”



“1” vs. “9”



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1. Bayesian Classification

- The likelihood model is no longer conjugate to the prior over the weight vector
- Approximate inference methods can be used (similar to Bayesian ranking) as the approximation is on a 1D-projection of each feature vector at each training example
- The message passing algorithm is highly efficient and requires no matrix inversion!

2. Sherman-Morrison

- Computationally efficient identity to compute the inverse after a rank-1 update
- There is also a rank- k version of the identity
- Used heavily in so-called LORA (“low rank”) adapters for training large language models!

**Introduction to
Probabilistic Machine
Learning**

*Unit 9 – Bayesian
Classification*

See you next week!